Stochastic Mathematics for Finance

Week 1 Assignment

Mathematical and Computational Case Studies in Stochastic Finance

Instructions.

- This assignment contains **4 finance-style case study questions**.
- Each question contains both a mathematical component and a computational component.
- Justify your mathematical answers clearly.
- For computational parts, submit the relevant Python code and a short explanation of your observations.
- You may use the Week 1 resources, but some parts require applying the ideas in a new setting.

Submission format.

Submit your work on GitHub in a clean repository containing:

- a `README.md` file with your name, roll number, and GitHub username;
- a single PDF named `solutions.pdf`;
- Python code files inside a folder named `code/`, named clearly as `q1.py`, `q2.py`, etc.;
- plots or generated outputs, if any, inside a folder named `outputs/`.

Question 1: Portfolio Loss Dependence

A risk manager is studying the one-day loss behaviour of a two-asset portfolio. Let

$$L_A = \{\text{asset } A \text{ suffers a loss tomorrow}\}, \qquad L_B = \{\text{asset } B \text{ suffers a loss tomorrow}\}.$$

Suppose

$$\mathbb{P}(L_A) = 0.10, \qquad \mathbb{P}(L_B) = 0.20, \qquad \mathbb{P}(L_A \mid L_B) = 0.35.$$

Define the indicator random variables I_A and I_B as follows:

$$I_A = \begin{cases} 1, & \text{if asset } A \text{ suffers a loss tomorrow,} \\ 0, & \text{otherwise,} \end{cases} \qquad I_B = \begin{cases} 1, & \text{if asset } B \text{ suffers a loss tomorrow,} \\ 0, & \text{otherwise.} \end{cases}$$

Now define

$$N = I_A + I_B.$$

Thus, N represents the number of assets suffering losses tomorrow.

(a) Compute $\mathbb{P}(L_A \cap L_B)$ and $\mathbb{P}(L_A \cup L_B)$.

(b) Are L_A and L_B independent? Justify mathematically.

(c) Compute $\mathbb{E}[N]$ and $\mathrm{Var}(N)$.

(d) Under a naive independence assumption, what would the joint loss probability be? By what factor does this underestimate or overestimate the true joint loss probability?

(e) Briefly explain why dependence between losses matters in portfolio risk management.

(f) Write Python code to simulate 10000 observations of (I_A, I_B) with the given probabilities.

(g) Estimate $\mathbb{P}(L_A \cap L_B)$, $\mathbb{P}(L_A \cup L_B)$, $\mathbb{E}[N]$, and $\mathrm{Var}(N)$ from the simulation. Compare these estimates with your theoretical answers.

Hint

For the simulation, first generate whether L_B occurs using

$$\mathbb{P}(L_B) = 0.20.$$

Then generate L_A conditionally. If L_B occurs, use

$$\mathbb{P}(L_A \mid L_B) = 0.35.$$

If L_B does not occur, use the law of total probability to first find $\mathbb{P}(L_A \mid L_B^c)$, and then use that value in the simulation.

Question 2: Credit Rating Migration

A credit risk analyst is studying the annual credit rating migration of a corporate bond. At the end of each year, the bond is classified into one of the following states:

$$S = \{0, 1, 2, 3\},$$

where

$$\begin{aligned} 0 &= \text{Investment Grade}, \\ 1 &= \text{Speculative Grade}, \\ 2 &= \text{Distressed}, \\ 3 &= \text{Default}. \end{aligned}$$

The transition probability matrix is

$$P = \begin{pmatrix} 0.60 & 0.25 & 0.05 & 0.10 \\ 0 & 0.50 & 0.50 & 0 \\ 0 & 0.50 & 0.50 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The rows and columns are ordered as $0, 1, 2, 3$. State 3 is absorbing. States 1 and 2 communicate only with each other, with probability 0.5 of staying in the same state and probability 0.5 of moving to the other state.

(a) Draw or describe the directed graph of the chain.

(b) Find all communicating classes and identify which of them are closed.

(c) Determine which states are recurrent and which states are transient.

(d) Is the chain irreducible? Is each communicating class aperiodic?

(e) Find the stationary distribution supported on each closed communicating class.

(f) Describe the full set of stationary distributions of the chain.

(g) Write Python code to simulate one path of the chain for 100 steps starting from state 0.

(h) Repeat the simulation 1000 times starting from state 0. Estimate the proportion of paths that are in state 3 at time 100, and the proportion of paths that are in the class $\{1, 2\}$ at time 100.

(i) Briefly interpret the closed communicating classes in the context of credit risk.

Hint

For a finite Markov chain, every state in a closed communicating class is recurrent. States not contained in any closed communicating class are transient.
For stationary distributions, handle each closed class separately.
If a closed class consists of a single absorbing state, then the stationary distribution supported on that class puts probability 1 on that absorbing state and probability 0 on every other state.
If a closed class contains two states, such as $\{1, 2\}$, then restrict the transition matrix to only those two states. In this question, the restricted matrix is

$$P_{\{1,2\}} = \begin{pmatrix} 0.50 & 0.50 \\ 0.50 & 0.50 \end{pmatrix}.$$

Now solve

$$(\pi_1, \pi_2)P_{\{1,2\}} = (\pi_1, \pi_2), \qquad \pi_1 + \pi_2 = 1.$$

After finding (π_1, π_2), assign probability 0 to all states outside the class $\{1, 2\}$.
Since this chain has more than one closed communicating class, the full set of stationary distributions is obtained by taking convex combinations of the stationary distributions supported on the closed classes.

Question 3: Inventory Random Walks

A market-making desk tracks inventory imbalance across d independent risk factors. The inventory vector is modelled as a simple symmetric random walk on

$$\mathbb{Z}^d.$$

The origin $0 \in \mathbb{Z}^d$ represents a perfectly neutral inventory position. At each step, the inventory changes by one unit in exactly one coordinate direction, either positively or negatively, with all $2d$ possibilities equally likely.

(a) In this financial interpretation, what does it mean for the origin to be recurrent? What does it mean for it to be transient?

(b) State the Markov chain recurrence criterion involving

$$\sum_{n=0}^{\infty} p_{00}^{(n)}.$$

(c) Explain why the simple symmetric random walk is recurrent for $d = 1$ and $d = 2$, but transient for $d = 3$.

(d) Write Python code to simulate simple symmetric random walks in dimensions $d = 1, 2, 3$, each starting from the origin.

(e) For each dimension, simulate 1000 paths of length 1000. Estimate the proportion of paths that return to the origin at least once after time 0.

(f) Repeat the experiment for path lengths $n = 100, 1000, 10000$. Briefly describe how the return behaviour changes with dimension and path length.

(g) Explain why simulation can suggest recurrence or transience, but cannot by itself prove either property.

Hint

For a Markov chain, a state i is recurrent if

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty,$$

and transient if

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} < \infty.$$

For a simple symmetric random walk on $\mathbb{Z}^d$, use the fact that $p_{00}^{(2n)}$ behaves like $\dfrac{C_d}{n^{d/2}}$ for large n, where $C_d > 0$ depends on d.
In the simulation, for each path, record whether the walk ever returns to the origin after time 0. The estimated return proportion is not a proof of recurrence or transience, because simulation only observes a finite number of paths for a finite time horizon.

Question 4: Market Regime Switching

A quantitative trading desk models the broad market environment using a weekly Markov chain. At the beginning of each week, the market is classified into one of four regimes:

$$S = \{0, 1, 2, 3\},$$

where

$$\begin{aligned} 0 &= \text{Bull}, \\ 1 &= \text{Neutral}, \\ 2 &= \text{Bear}, \\ 3 &= \text{Crisis}. \end{aligned}$$

The weekly transition probability matrix is

$$P = \begin{pmatrix} 0.60 & 0.30 & 0.10 & 0 \\ 0.20 & 0.50 & 0.20 & 0.10 \\ 0.10 & 0.30 & 0.40 & 0.20 \\ 0.05 & 0.15 & 0.30 & 0.50 \end{pmatrix}.$$

The rows and columns are ordered as $0, 1, 2, 3$.
The expected weekly portfolio returns in the four regimes are

$$r_0 = 0.012, \qquad r_1 = 0.004, \qquad r_2 = -0.006, \qquad r_3 = -0.025.$$

(a) Is the Markov chain irreducible and aperiodic? Justify briefly.

(b) Find the stationary distribution

$$\pi = (\pi_0, \pi_1, \pi_2, \pi_3)$$

by solving

$$\pi P = \pi, \qquad \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1.$$

You may solve this either by hand or using Python, but explain your method.

(c) Interpret π_i in the context of market regimes.

(d) Using the stationary distribution, compute the long-run average weekly portfolio return.

(e) Compute the long-run fraction of weeks spent in unfavourable regimes, where Bear and Crisis are considered unfavourable.

(f) Write Python code to simulate one path of the Markov chain for 10000 weeks, starting from the Bull regime.

(g) Estimate the empirical fraction of time spent in each regime and compare it with the stationary distribution.

(h) Using the simulated regime path, compute the empirical average weekly portfolio return and compare it with the theoretical long-run average return.

(i) Plot the cumulative portfolio return over time, defined by

$$C_n = \sum_{t=1}^{n} r_{X_t}.$$

(j) Briefly explain why simulation is useful even when the stationary distribution can be computed theoretically.

Hint

The stationary distribution gives the long-run fraction of time spent in each regime. Once you have π, the long-run average weekly return is

$$\sum_{i=0}^{3} \pi_i r_i.$$

The long-run fraction of unfavourable weeks is

$$\pi_2 + \pi_3.$$

For simulation, if the current state is i, sample the next state from $\{0, 1, 2, 3\}$ using the probabilities in row i of P. As the number of simulated weeks becomes large, the empirical regime frequencies should become close to the stationary distribution.